

Braneworld

The three physical dimensions plus time is known as Braneworld. String theory predicts any number of extra dimensions.

Space tourism

The Space Investment Summit 6 was held on May 27, to discuss the future prospects of space tourism

Fold the data as well

Scientists at Swinburne University showcased plans of DVDs that store data in 5 dimensions, allowing 2000 movies on one disc

bizarre predictions. Firstly, it suggested that massless particles could exist – and since even photons supposedly have mass, that's a disturbing thought. Secondly, it proposes that our universe has not four, but eleven distinct dimensions – a brain-melting thought on its own – which opens up the door for multiple universes.

In fact, these multiple dimensions and universes are essential for string theory to truly explain gravity.

'Branes and 'verses

It's hard enough to think of the universe as having four dimensions, much less eleven. String theorists suggest that we don't see these dimensions because they're either too small or too large to be observed.

To understand the first idea, imagine that you're a super-massive being, looking down at the Milky Way. To you, Earth is just a point, if that; it's one-dimensional. To us regular

shiny new LCD monitor.

The game's characters seem to be moving in three dimensions – they're hiding behind objects, rushing at you, and running away from you. Of course, they're not really moving in three dimensions – they're trapped inside the two dimensions of your screen. What the game's characters see as three-dimensional space, you see as an optical illusion. This might be what's happening to our universe – our four dimensions might just be someone else's optical illusion. With the possibility of extra dimensions, string theory says that it's possible that one string can stretch to form a huge membrane – or brane, in short – that encompasses our entire universe. So while we think we've got it made with four whole dimensions, we're still trapped inside a brane. And if one string can turn into a brane, so can others – each with its own universe. And just like the characters from your screen can't jump out into the world, we can't jump from brane to brane, even if they're just a millimetre apart.

And in all this weird wondrousness, we might have an explanation for gravity – or more specifically, why it's so weak.

Leaky universes

It might not be that gravity is a weak force, after all – it might be just as strong as electromagnetism, but something might be robbing it of its strength.

String theory talks of two types of strings. The first – the open-ended string – is tied to our brane by both its ends. If it wants to travel to another universe, too bad – it can't. The second, however, is a closed loop, like a rubber-band. It isn't confined to just one brane, it comes and goes as it pleases. These closed loops make up gravitons – the missing particle in the Standard Model. When bodies exchange gravitons, they are attracted to each other, producing gravity. But since gravitons are closed-loop strings, they're escaping into

BIG BANGING BRANES

If we are to accept that we're trapped in a four-dimensional brane in a larger universe with more dimensions, we might also have an explanation for the Big Bang. With all these branes floating about, it might be that another brane collided with ours, spewing all of its contents into ours.

But scientists aren't comfortable

other universes, weakening its influence in our universe.

And so, with the flighty graviton, string theory plugs a major hole in the Standard Model – it plausibly ties together all of nature's forces, becoming the Theory of Everything. But is this too good to be true?

The great test

While string theory seems plausible, it's only plausible in the way that the existence of God is plausible. There's no way to prove either notion wrong – but that doesn't mean they are false. On the other hand, there's no way to prove them, either, so they aren't really true, either. According to string theory's critics, if it's

with this idea. First, like the rest of string theory, there's no way to prove this. Second, it implies that other universes existed before ours, and others before them. In essence, the universe doesn't really have an origin. And a universe without an origin is a universe that doesn't make sense at all.

we know today – also called M-theory – but it'll always have an embarrassingly fuzzy past.

A growing number of scientists, however, have faith in the theory – to them, it just makes too much sense to be wrong. Still, they need to figure out a way to test it, because that faith won't last forever. And that test might just show up in an experiment we already have high expectations from.

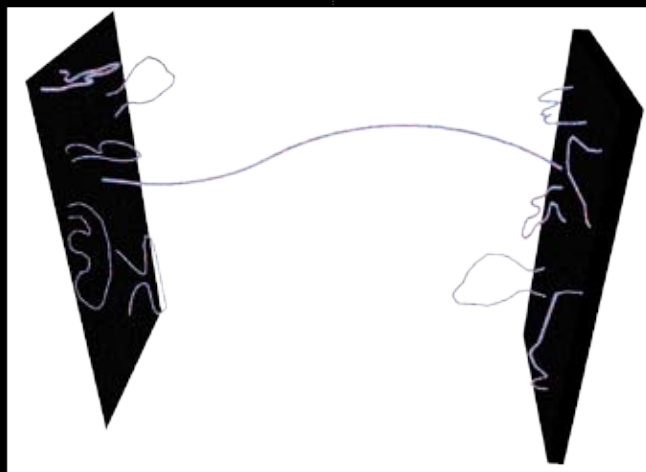
If string theory is correct, there's a small chance (0.5 percent, to be precise) that the Large Hadron Collider (LHC) will show us the proof. In its atom-smashing frenzy, the LHC might just manage to catch a glimpse of a graviton before it escapes to another dimension. It's a small probability, but it's



String theorists imagine that the universe's extra dimensions might be squished into tiny globes like this...

humans, though, it has a full three dimensions – we move back and forth, up and down, and left and right. It's as if our three dimensions have been squished into a single one of your dimensions. In the same way, the extra dimensions that string theory predicts might just be squished into tiny, subatomic spaces.

The second, more popular idea, is that these extra dimensions are so large that we can't notice them – just like we can't notice that the world is round from where we're standing. Imagine now that you're playing a game on your



...or stretched out into massive membranes that hold whole universes

not testable, it's not science. Case closed. It became even worse for string theory when it came out in five plausible forms. If there's supposed to be a single theory that explains the universe, how can it come in five flavours, each of which makes sense? It was a while before these five theories were integrated into the string theory

better than nothing. Even if the LHC doesn't spot a graviton, it'll do nothing to harm the theory.

For now, however, if we are to believe in string theory, we must remain content with the hope that there might some day be a way to prove or disprove it, and that there really is a single way to describe all the forces in the universe. **d**